

TIMELINE

- c.1500BCE** Polynesians start to colonize most of the islands of the Pacific Ocean, crossing it in large, double-hulled sailing canoes. They reach Hawaii and New Zealand by 1000 CE.
- c.985CE** Viking mariners reach the coast of Newfoundland in North America after being driven southwest by a storm while sailing to Greenland from Iceland.
- 1405–33** Chinese admiral Zheng He explores the Indian Ocean with his fleet.
- 1492** Genoese merchant Christopher Columbus uses the trade winds to cross the Atlantic Ocean in search of a sea route to China, but discovers the West Indies instead.
- 1519–21** Portuguese navigator Ferdinand Magellan is sent by Spain to find a sea route to the East Indies via the Pacific. After he is killed in the Philippines his crew sails home across the Indian Ocean, and completes the first voyage around the world.
- 1735** British physicist George Hadley explains how the Earth's rotation affects the oceanic trade winds. His name is given to the Hadley cells, which are part of the pattern of global air circulation.
- 1759** British clockmaker John Harrison perfects an accurate, reliable marine chronometer that can be used to determine longitude at sea.
- 1768–79** British sailor Captain James Cook undertakes three voyages of exploration, mainly in the Pacific and Southern Ocean.
- 1770** Captain Cook and his crew discover the Great Barrier Reef on the eastern coast of Australia when their ship the *Endeavour* is almost wrecked on it.
- 1770** American inventor and statesman Benjamin Franklin publishes the first map of the Gulf Stream.
- 1805** British Admiral Francis Beaufort devises a scale for measuring and describing wind speed at sea.
- 1819** Swiss chemist Alexander Marcet discovers that the basic chemistry of seawater is the same throughout the world.
- 1831–36** British naturalist Charles Darwin conducts some of the earliest oceanographic research during the voyage of the HMS *Beagle*.
- 1835** French physicist Gustave-Gaspard de Coriolis publishes a paper describing how ocean currents and moving air are affected by the spinning Earth. This becomes known as the Coriolis effect.
- 1839–43** British navigator James Clark Ross commands an expedition to Antarctica, discovering the Ross Sea and the Ross Ice Shelf.
- 1842** Darwin publishes his theory of the structure and formation of coral reefs and atolls. However, he is not proved right until the 1940s.
- 1843** British naturalist Edward Forbes says that life cannot exist below 1,800 ft (550 m) in the oceans, starting a debate about the presence of an azoic (lifeless) zone.
- 1868–69** During dredging operations from HMS *Lightning* and *Porcupine*, Scottish naturalist Wyville Thomson discovers oceanic life as deep as 14,403 ft (4,390 m). This disproves Edward Forbes' theory of a deep-ocean lifeless zone.
- 1869** The Suez Canal between the Mediterranean and the Red Sea is completed after 11 years' work, but at the cost of at least 125,000 lives.
- 1872** British scientist Sir William Thomson (Lord Kelvin) invents a sounding machine that uses wire cable to measure ocean depths. This is more accurate than older sounding methods using weighted hemp rope.
- 1872–76** HMS *Challenger* makes the first thorough scientific exploration of the oceans. The ship travels 68,910 miles (110,900 km) on its four year voyage.
- 1874–75** Charles Sigsbee of the US Navy pioneers new methods of mapping the ocean floor when surveying the Gulf of Mexico.
- 1882** The first vessel specifically built as an oceanographic research vessel begins operations—the US Fish Commission steamer *Albatross*.
- 1883** The volcano Krakatau erupts between Java and Sumatra, causing tsunamis that kill 36,000 people.
- 1893–96** Norwegian explorer Fridtjof Nansen's ship *Fram* drifts over the central Arctic Ocean while locked in pack ice, demonstrating that the polar ice moves.
- 1897** The first offshore oil rig is built off the coast of California, just 295 ft (90 m) from the shore.
- 1900** The coastal town of Galveston, Texas, is destroyed by a hurricane. The storm surge and resulting floods kill more than 8,000 people.
- 1903** Norwegian explorer Roald Amundsen leads the first expedition to traverse the Northwest Passage through the Arctic successfully.
- 1912** The steamship *Titanic* sinks after colliding with an iceberg that has drifted 435 miles (700 km) southeast of Newfoundland; 1,500 people die in the mishap.
- 1914** The Panama Canal between the Caribbean and the Pacific Ocean is opened after a lengthy construction project.
- 1915** German meteorologist Alfred Wegener publishes his theory of continental drift, suggesting that all the continents were once joined together in a "supercontinent," one that he later calls Pangaea.
- 1915** British explorer Ernest Shackleton's ship *Endurance* drifts 800 miles (1,300 km) while locked in the pack ice of the Antarctic Weddell Sea before being crushed and destroyed.
- 1919** French scientists succeed in using an echo sounder (sonar) device to make the first depth measurements.
- 1924** Russian biochemist Aleksander Oparin suggests that life on Earth could have originated in the oceans, as simple substances linked up to form the first complex molecules essential to life.
- 1925–27** German survey ship *Meteor* uses sonar to produce a profile of the Mid-Atlantic Ridge.
- 1934** US naturalist Charles Beebe and engineer Otis Barton make a world record descent of 3,028 ft (923 m) in a simple submersible called a bathysphere.
- 1943** The first self-contained underwater breathing apparatus (SCUBA) is developed by French diver Jacques Cousteau and engineer Emile Gagnan.
- 1948** US oceanographer Henry Melson Stommel publishes a paper explaining the workings of the Gulf Stream and the ocean circulation that redistributes heat around the world.